

1. ISMS Overview

1.1 Purpose of an ISMS

Paste this **exact text**:

An Information Security Management System (ISMS) is a systematic approach for managing information security within an organization. It provides a structured way to identify and manage information-security risks, define responsibilities, apply appropriate security practices, and protect organizational information and the systems that support it.

The purpose of an ISMS is to ensure that information-security activities are managed in a consistent and organized manner. It brings together people, processes, policies, and security controls to support the protection of organizational information.

In the SecureWise ISMS Cyber Security Challenge, the ISMS concept is introduced through practical workplace situations. The course focuses on helping employees understand how their everyday security decisions and behavior contribute to protecting organizational information.

1.2 Continual Improvement

Paste this immediately below:

Information security is not a one-time activity. Security risks, technologies, systems, and working practices can change over time. Therefore, an ISMS requires continual review and improvement.

Continual improvement involves reviewing existing security practices, identifying weaknesses or changes in information-security risks, taking appropriate improvement actions, and reviewing the results. This helps an organization keep its information-security practices relevant as its environment and risks change.

In the SecureWise course, continual improvement is reflected through learning, practical decision-making, feedback, assessment, and review of learner performance.

2. CIA Triad

2.1 Confidentiality

Confidentiality means protecting information from unauthorized access or disclosure. Information should only be accessible to people who are authorized to use it. Maintaining confidentiality helps prevent sensitive organizational information from being exposed to unauthorized individuals.

For example, confidential employee information should only be accessible to authorized personnel who require it for their work.

2.2 Integrity

Integrity means protecting information from unauthorized or improper modification. Information should remain accurate, complete, and trustworthy so that it can be relied upon for business activities and decisions.

For example, an unauthorized person should not be able to modify important employee or business records.

2.3 Availability

Availability means ensuring that information and supporting systems are accessible to authorized users when they are required. A security approach should therefore consider whether important information and systems can remain available for legitimate business use.

For example, an authorized employee should be able to access an important business system when it is required for their work.

2.4 Importance of the CIA Triad

Confidentiality, integrity, and availability represent three important information-security requirements. An organization needs to consider all three when protecting its information and systems.

In the SecureWise course, these concepts form the foundation of Mission 1, “Protect the Information,” where learners are expected to recognize confidentiality, integrity, and availability needs and apply them while classifying information.

3. Information Security Risk

Information-security risk can be understood by considering the asset that needs protection, the threats that may affect it, the vulnerabilities that can be exploited, the likelihood of an event occurring, and the resulting impact.

The relationship used for this project is:

Asset → Threat → Vulnerability → Likelihood → Impact

An **asset** is something valuable that needs to be protected, such as organizational information, an account, a document, or a system.

A **threat** is a potential source of harm that may affect an asset. Examples include phishing attacks, unauthorized access, malware, or accidental disclosure.

A **vulnerability** is a weakness that a threat may exploit. Vulnerabilities can exist in systems, processes, or human behavior.

Likelihood represents how possible it is that a particular security event will occur. **Impact** represents the consequences if the event occurs.

For example, a company email account can be considered an asset. A phishing attack can be a threat, while unsafe handling of a suspicious link can represent a vulnerability. The likelihood describes the possibility of the attack succeeding, while the impact describes the potential consequences of a successful compromise.

Understanding this relationship helps employees recognize how everyday security decisions can contribute to information-security risk.

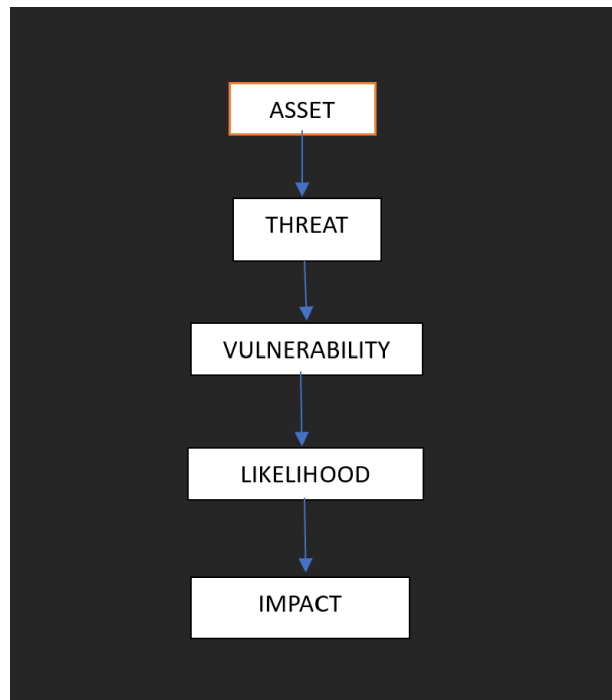


Figure 1: Basic Information-Security Risk Relationship

4. Human Security

4.1 Employee Responsibilities

Employees play an important role in protecting organizational information. Information security is not limited to technical systems and security tools; employees also need to follow appropriate security practices in their daily work.

Relevant employee responsibilities for the SecureWise course include recognizing suspicious emails and links, following secure password and authentication practices, handling information appropriately, and reporting suspected security incidents through the correct channel.

Employees should also follow the security procedures and responsibilities defined by their organization when handling information, systems, accounts, and security-related events.

4.2 Security Behavior

Security behavior refers to the actions and decisions employees make while performing their normal work. Safe security behavior means recognizing potential security risks and choosing actions that reduce the possibility of unauthorized access, information disclosure, modification, or disruption.

For example, an employee who receives a suspicious email should not automatically open its links or attachments. The employee should recognize the warning signs and follow the appropriate safe response.

The SecureWise course uses workplace scenarios to allow learners to practice these decisions. Learners are presented with a situation, asked to make a decision, and given immediate feedback explaining why the selected action is correct or unsafe.

4.3 Role in the SecureWise Course

Human security behavior is represented across multiple missions in the course. Mission 1 addresses appropriate handling and classification of information. Mission 2 focuses on recognizing phishing indicators and selecting a safe response. Mission 3 addresses secure password and authentication practices. Mission 4 focuses on recognizing security incidents and following the correct reporting path.

These activities are intended to connect information-security concepts with practical employee decisions in realistic workplace situations.

5. SCORM and SCORM 1.2

5.1 SCORM Overview

SCORM (Sharable Content Object Reference Model) provides a standard approach for packaged e-learning content to launch from and communicate with a Learning Management System (LMS). For the SecureWise project, the browser-based learning course will be packaged as a SCORM 1.2 single SCO so that it can be imported and launched from an LMS.

The SCORM runtime allows the course to communicate learner information to the LMS, including lesson status, score, current location, and saved learner state. This communication is required for the course to support accurate tracking and resume behavior.

5.2 SCORM 1.2 Runtime Lifecycle

The planned SCORM 1.2 runtime lifecycle begins when the learner launches the course from the LMS. The course first locates the LMS-provided API and initializes the communication session. It then reads existing learner information such as lesson status, location, score, and suspend data.

If valid saved data is available, the course restores the learner's saved state. As the learner completes activities and their progress or score changes, the course updates the required SCORM values. Meaningful progress is committed to the LMS so that it can be retained.

When an unfinished course is exited, the appropriate suspend behavior is used so that the learner can resume later. When the session ends, the course terminates the SCORM communication using the finish operation. Initialization and termination should occur only once during a session, and important runtime operations should have their results and errors checked.

5.3 SCORM Runtime Sequence

The runtime sequence for the SecureWise course is:

Learner Launch → Find LMS API → Initialize → Read Saved Data → Restore Learner State → Learner Activity → Update Score/Progress → Set Values → Commit → Exit → Finish

This sequence allows the course to communicate the learner's progress and result to the LMS and supports the required resume behavior.

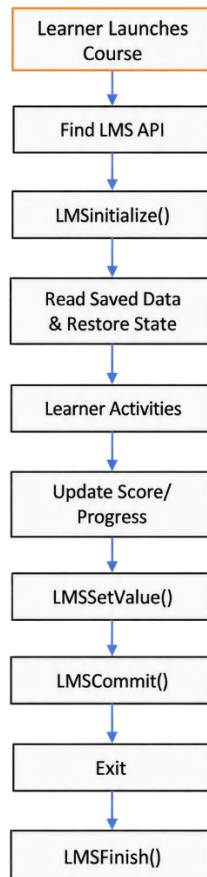


Figure 2: SCORM 1.2 Runtime Sequence

5.4 SCORM 1.2 Data Mapping

SCORM 1.2 Element	Purpose in the Course
cmi.core.lesson_status	Records incomplete, completed, passed, or failed status
cmi.core.score.raw	Records the learner's current or final score
cmi.core.score.min	Defines the minimum score as 0
cmi.core.score.max	Defines the maximum score as 100
cmi.core.lesson_location	Records the current mission or resume location
cmi.suspend_data	Stores the serialized learner state
cmi.core.exit	Uses suspend behavior when the course is unfinished

5.5 SCORM State and Resume

The SecureWise course needs to maintain a learner state that can be saved and restored between sessions. The saved state should contain the information required to continue the learner's actual

course experience, including the current mission, completed missions, mission scores, answers, badges, and assessment state.

The serialized state must be versioned and kept within the practical SCORM 1.2 suspend-data limit. Invalid or unknown saved-state versions should be handled safely rather than being used without validation.

6. Course UI State vs LMS Tracking State

6.1 Course UI State

Course UI state refers to the information that the course application maintains while the learner is using the course.

It represents what is currently happening inside the course. For example, it can include the learner's current mission, completed missions, answers, scores, badges, and assessment progress.

The UI state helps the course display the correct content and maintain the learner's current experience.

For example, if a learner has completed Mission 1 and is currently working on Mission 2, the course UI state keeps track of this information so that the application can display the correct mission and progress.

6.2 LMS Tracking State

LMS tracking state refers to the information that the course communicates to the Learning Management System through SCORM 1.2.

The LMS is mainly concerned with tracking important information about the learner's course progress and result.

For the SecureWise course, important SCORM 1.2 tracking values include:

cmi.core.lesson_status – Records the course status such as incomplete, completed, passed, or failed.

cmi.core.score.raw – Records the learner's score.

cmi.core.lesson_location – Records the learner's current mission or resume location.

cmi.suspend_data – Stores the saved learner state required for resuming the course.

6.3 Difference Between the Two

The main difference is simple:

Course UI State = What the course application knows about the learner.

LMS Tracking State = What the LMS needs to know about the learner.

The course UI state is used to manage the learner's detailed experience inside the application. The LMS tracking state is used to communicate important progress, score, status, and resume information to the LMS through SCORM.

The two states are connected through the SCORM communication layer.

Course UI State → SCORM Communication → LMS Tracking State

6.4 Simple Example

Suppose a learner completes Mission 2 and receives a score of 80.

Inside the course, the UI state can record:

Current Mission: Mission 2

Mission Status: Completed

Score: 80

Answer: Selected answer

Badge: Earned

The course then communicates the required tracking information to the LMS through SCORM.

The LMS can record:

Score: 80

Status: Completed or Passed, according to the course result

Location: Mission 2

Saved State: Stored through suspend data

Therefore, the **UI state manages the detailed learner experience**, while the **LMS tracking state records the information required by the LMS**.

6.5 Resume

Resume functionality connects these two states.

When a learner leaves an unfinished course, the required learner state can be saved through SCORM. When the learner launches the course again, the saved information is read and used to restore the learner's previous course state.

For the SecureWise project, resume should restore the learner's actual state rather than showing only a progress bar. The handbook specifically requires the restored state to include information such as answers, mission state, score, and badges.

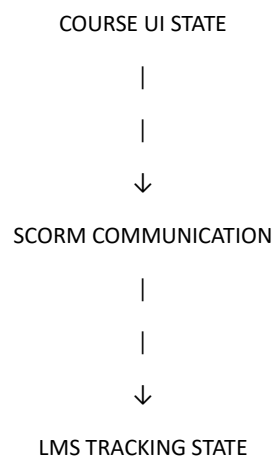


Figure 3: Course UI State and LMS Tracking State

Course UI State:

Mission • Answers • Scores • Badges • Assessment State

LMS Tracking State:

Status • Score • Location • Suspend Data

7. Technical Risks That Can Cause SCORM Course Failure

7.1 LMS API Not Found

A SCORM course needs to communicate with the LMS through the SCORM API. If the course cannot locate the LMS-provided API, SCORM communication cannot be established and tracking operations may fail.

7.2 SCORM Initialization Failure

The course must successfully initialize the SCORM communication session before reading or writing tracking information. If `LMSInitialize()` fails or returns an unsuccessful result, the course may not be able to communicate learner progress and other tracking information correctly.

7.3 Incorrect Tracking Data

Incorrect mapping or updating of SCORM values can result in inaccurate learner records. For example, an incorrect score, lesson status, or lesson location may cause the LMS to display information that does not match the learner's actual course progress.

7.4 Invalid or Lost Suspend Data

Suspend data is important for saving learner state between sessions. If the serialized learner state is malformed, exceeds the practical SCORM 1.2 limit, or cannot be restored correctly, the learner may lose progress or resume at an incorrect state.

7.5 Failure to Commit or Finish the Session

If important progress is not committed to the LMS, the latest learner information may not be saved correctly. Similarly, incorrect termination of the SCORM session can cause tracking or completion information to be handled incorrectly.

For this reason, the course should commit meaningful progress and properly terminate the SCORM communication session.

7.6 Risk Summary

The five major technical risks identified for the SecureWise SCORM course are:

1. LMS API cannot be located.
2. SCORM initialization fails.
3. Tracking data is incorrect or incorrectly mapped.
4. Suspend data is invalid, lost, or exceeds the practical limit.

5. Progress is not committed or the SCORM session is not properly terminated.

These risks will be considered during the later development, SCORM integration, packaging, and testing stages of the project. The handbook specifically requires API initialization, saved-progress restoration, score/status accuracy, commit handling, and proper session termination to be verified during SCORM integration and testing

8. Proposed Learning Experience and Target Audience

8.1 Proposed Learning Experience

The SecureWise ISMS Cyber Security Challenge is proposed as a browser-based gamified learning experience designed to help learners understand basic information-security concepts through practical workplace situations.

The course will be organized into five missions. Each mission will introduce a specific information-security topic and provide learners with an opportunity to apply the concept through interactive activities and decision-based scenarios.

The proposed course structure is:

Mission 1 – Protect the Information

Focuses on confidentiality, integrity, availability, and appropriate information classification.

Mission 2 – Spot the Phish

Focuses on recognizing phishing indicators and selecting an appropriate response to suspicious messages.

Mission 3 – Lock It Down

Focuses on secure password practices and multi-factor authentication.

Mission 4 – Report It Right

Focuses on recognizing security incidents and following the correct reporting process.

Mission 5 – Final Assessment

Provides a final assessment covering the concepts introduced throughout the course.

The learning experience will use short explanations, workplace scenarios, questions, immediate feedback, scoring, progress tracking, badges, and retry opportunities. This approach is intended to connect information-security concepts with practical decisions that learners may encounter in their work.

The completed course is expected to have a maximum score of 100, with an 80 percent passing threshold as defined in the project requirements.

8.2 Target Audience

The primary target audience is employees who need to understand basic information-security responsibilities and secure workplace behavior.

The course is intended for learners who may not have an advanced technical or cybersecurity background. Therefore, the learning experience will use simple explanations and realistic workplace situations rather than relying mainly on technical terminology.

The scenarios will focus on common employee security decisions, such as handling information appropriately, identifying suspicious emails, using secure authentication practices, and reporting suspected security incidents.

All scenario people, organizations, email addresses, and other example data used in the course will be fictional, in accordance with the project requirements.

8.3 Learning Approach

The proposed learning approach follows a simple cycle:

Learn → Practice → Decide → Receive Feedback → Earn Score → Progress

Learners will first understand the concept, then apply it to a practical situation. Their decision will produce immediate feedback, allowing them to understand why an action is appropriate or inappropriate.

The gamified elements such as score, progress, badges, and retry opportunities are intended to make the learning experience interactive while maintaining the focus on information-security awareness.

8.4 Expected Learner Outcome

By completing the course, learners should have a basic understanding of information-security principles and should be able to apply them to common workplace situations.

The course will particularly emphasize responsible information handling, recognition of phishing attempts, secure authentication practices, and appropriate security-incident reporting.



Figure 4: Proposed Learning Experience

9. References

1. SecureWise, "Phase 2 Intern Implementation Handbook," SecureWise Internal Training Document, Day 1: Understand ISMS and SCORM.
2. ADL, "SCORM 1.2 Documentation," Advanced Distributed Learning Initiative. Reference material for SCORM 1.2 runtime communication, learner tracking, and course packaging.

3. ISO/IEC, "ISO/IEC 27001 — Information Security Management Systems." Reference material for understanding the purpose and management of information security within an organization.
4. NIST, "Cybersecurity Framework (CSF) 2.0," National Institute of Standards and Technology. Reference material for cybersecurity risk and information-security practices.

Note on Sources

The SecureWise handbook was used as the primary project-specific reference for the required Day 1 topics, terminology, SCORM 1.2 requirements, course structure, and project deliverables. External references were used to support the general understanding of information-security management, cybersecurity risk, and SCORM concepts.